

SKINNER'S SCIENTIFIC SPECTRUM

“The biological sciences, as the study of both nature and man, cannot be underestimated in power. Their domain is us.

- NEO SKINNER

MATHEMATICS

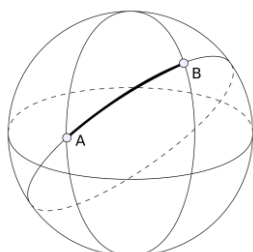
A FIVE-SIDED SQUARE: IGNORING EUCLID

The standard laws of geometry were outlined by a fine fellow called Euclid who came from the ancient Greek city of Alexandria in somewhere around 300 BCE. These laws centred around five axioms, which consist of basic statements assumed to be true, the first of these simply states: “Given two points there is one straight line that joins them.” (The Editors of Encyclopedia Britannica, 2024) What happens if we construct a system that doesn’t follow this rule?



The Fine Fellow in Question; LTR: Euclid's the Elements, an Artist's Interpretation of Euclid

I know what you are thinking, how can we defy Euclid? Simple: when a straight line is not straight. Picture, for a moment, a solid sphere in your head. Draw two points on your sphere and join them with a line. This line may at first seem straight, but think again, that line is actually a 3D curve. This is the nature of spheres, the shortest distance between two points is an arc of what is called a great circle, and they can be found in our lives as the routes aeroplanes take across our approximately spherical planet.



Taking the Shortest Route; LTR: An Arc of a Great Circle on a Sphere, the Longest Commercial Flight Route

A spherical plane is one of the alternative geometries to that of Euclid, who requires a flat plane, and it leads to some interesting consequences. If you once again try to conjure a sphere in your mind and this time attempt to draw a square, you will quite quickly find that your

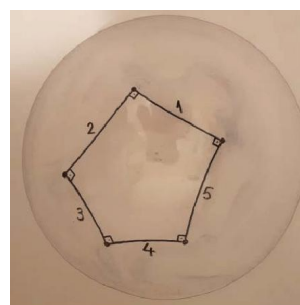
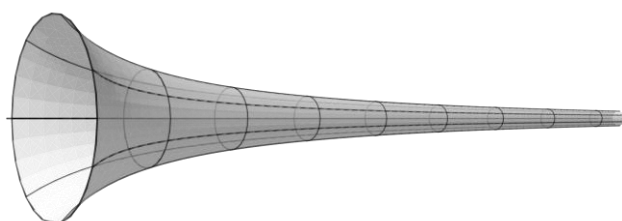
‘square’ instead has three sides joined by right angles instead of the usual four! This is because as the sphere warps the straight lines you draw onto it, it also increases the internal angles of the shapes formed – stretching the 60-degree angles of an equilateral triangle to instead be 90. Weird.

This stretching can be explained by the type of curvature exhibited by the plane, for a sphere, any two perpendicular lines will both bulge outwards, with bulge κ , and thus a sphere has a constant positive curvature, K ($+\kappa_1 \times +\kappa_2 = +K$). A positive curvature will always ‘inflate’ angles as would an inflating balloon.



Deep Breaths; Angles ‘Inflating’ on the Surface of a Balloon (However Faint!)

Now, let us consider a different type of plane which breaks Euclid’s neat axioms: Gabriel’s horn is the plane formed when the shape of a reciprocal ($\frac{1}{x}$) graph is rotated over the x - or y -axis and has the curious property of constant negative curvature ($-\kappa_1 \times +\kappa_2 = -K$) for the external section. A plane formed this way is called hyperbolic, as another name for the reciprocal graph shape is a hyperbola.



Curiouser and Curiouser; LTR: Gabriel’s Horn, a Hyperbolic Pentagon (Spoiler!)

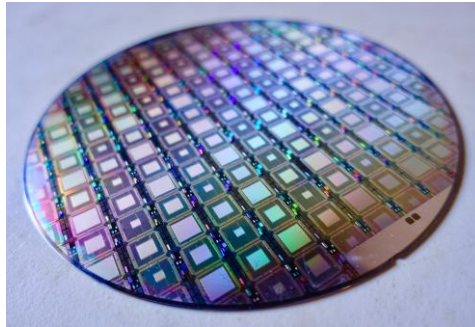
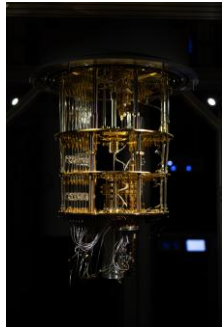
A constant negative curvature will decrease the angles of shapes drawn on the plane surface and our familiar triangle will now have angles which are all less than our 60-degree stronghold. But what for the 90-degree angle? Instead of four sides in the Euclidean world and three in the spherical plane, right angles now form a five-sided shape - a pentagon!

If you defy Euclid, the results may seem strange - but only from our point of view. The world we view is inherently Euclidean, yet it could have - just as easily - been spherical or hyperbolic. Indeed, it is believed that our universe may be curved in accordance with the hyperbolic nature of a horse saddle - the measurements required are just too large for us to know for certain.

PHYSICS

FROM 0 TO... 1

Computing in its rawest form has existed for just under 5 000 years (a), beginning with the Babylonian abaci and developing to the high-power supercomputer networks of today (NWE et al., 2023). But, whilst data zips beneath us through network cables and flies over our heads as electromagnetic waves, there is another computing revolution on the horizon. Quantum computing.



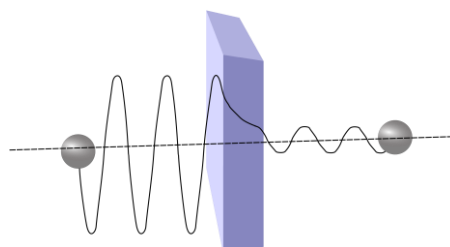
Move Over Abacus; LTR: A Quantum Computer, a Silicon Wafer of Quantum Processors

At the heart of quantum computing are qubits. Qubits are complicated to understand, so let us build a qubit to understand how they work - never fear, you will not need millions of pounds of equipment, we will create an imaginary qubit!

A qubit sounds surprisingly simple to create - more on the 'sounds' later - merely consisting of two superconductors separated by an insulative barrier. This arrangement is known as a Josephson junction, and we would normally think of this construction as something similar to a capacitor - whereby two charged plates are separated by an insulator - which would hold and release charge. However, Josephson junctions do not hold charge, they allow for it to flow.

How? Quantum tunnelling: what I have failed to mention so far is that the insulator in these junctions is about as thick as a singular glucose molecule ($30 \text{ \AA}$ (Newrock, 1997) - hence the 'sounds' earlier), which means that electrons are able to 'jump' across the boundary. This is because electrons do not behave as you would expect.

Electrons can be represented using things called probability waves, where each unit of space has an associated probability. Together these probabilities describe where it is likely to find an electron. As the nature of electrons is highly uncertain, it allows them to tunnel through fine boundaries where probabilities may be low but still possible.



Quantum Tunnelling in Action

Quantum tunnelling allows for charge to be carried across the barrier of the Josephson junction and this is what determines the state of the junction at any point in time. By firing microwave photons (light of the microwave frequencies) at the junction, the electrons can be fixed in place or moved, with the resulting value also determined through the process.

Another key feature of the quantum world is superposition. As the electrons within a Josephson junction ‘jump’ between the two superconductors, the junction gains an associated probability function. This function is similar to the electron waves we saw previously, but not the same. We can define this function as the sum of the electron probability waves, providing the likelihood that the Josephson junction has a positive or negative charge density, where each of the superconductors are assigned to be either positive or negative - if there is a high charge density on the negative plate, a temporary current can flow.

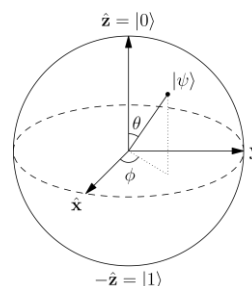
$$f(t) = P(e_1, t, x^+) + P(e_1, t, x^-) + P(e_2, t, x^+) + \cdots + P(e_n, t, x^Q)$$

A Representation of the Probability Function of a Josephson Junction

This probability function represents the superposition of the electron waves. In a superposition, we consider every possible value of the probability function as actually occurring - but hold on, does not the probability determine which value occurs? Not quite. We say that the probabilities only ‘lock in’ when we make a measurement of the system and thus, whilst our predictions may accurately describe the system we observe, they cannot describe the unobserved state. This is a tricky concept to grapple with and raises the transcending philosophical debate within quantum mechanics of reality - perhaps I will cover this at some point... But, for now, just take this paragraph at face value and read it as many times as you need!

The Josephson junction forms the central part of a qubit, and it is the probability function that makes them so useful. When the probability function is entirely positive, all of the electron waves lie in the positive superconductor - representing a 0. In the opposing case when the function is entirely negative, a 1 is the result. Our traditional binary operation is restored in these two *edge* cases.

I label them as edge cases as the function will normally lie somewhere between the two states, on what is known as a Bloch sphere. This is because the function provides a complex output which can be represented as a vector inside a sphere - allowing for complex computations using multiple qubits, we have made a quantum computer!



A Bloch Sphere

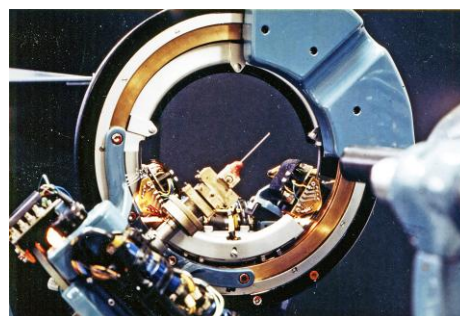
CHEMISTRY

GROWING CRYSTALS FOR... ANALYSIS?

The opening of Walt Disney World, Florida, on the 1st of October 1971 may seem an exciting and ambitious prospect. But, in fact, it is not the exciting event we are interested in, for this story begins one month later on the 20th of November 1971 with the publication of a landmark paper in the *British Medical Journal* by a 61-year-old Dorothy Hodgkin.

In 1971, Hodgkin was an X-ray crystallographer operating out of Wolfson College, Oxford, and had just published the structure of insulin - the fourth and final of her great discoveries. Insulin is a biological molecule made of 51 amino acids with a total mass of 5 794 amu and is essential within our bodies as the trigger for glucose storage as glycogen (NCBI, 2012). Insulin's complexity meant that it took 50 years to uncover its structure following its discovery in 1921 and was only made possible through an amazing technique called X-ray crystallography.

In X-ray crystallography, high-power X-rays are fired at a crystal lattice sample in a machine known as a diffractometer which rotates the sample whilst collecting data as an X-ray image formed using a CCD (charge-coupled device) or a photographic plate.

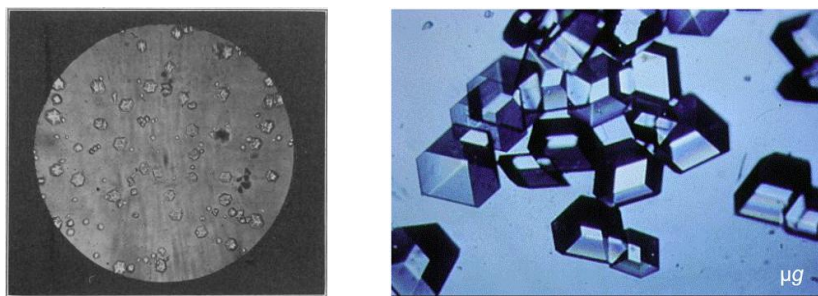


An X-Ray Diffractometer (the Hilger and Watts Y290); LTR: The Full Device, a Sample in the Sample Holder

This technique requires a crystalline sample - a compound in its crystal lattice form - which will interact with the incident X-rays, 'diffracting' them in differing directions. By 'diffraction' we simply mean that X-rays are rotated by the sample through a measurable angle to produce multiple distinct rays.

Most of the difficulty with this technique comes with obtaining a crystalline sample to perform measurements with. In the case of Hodgkin's earlier successes, smaller molecules such as the steroid cholesteryl iodide and the drug penicillin were used with relatively small molecular sizes - this makes them both easier to crystallize and to predict the structures for. But insulin is different, it is both large and complex - making it difficult to crystallize and to analyse.

In 1926, the first of the two problems was solved - insulin was crystallized by John Jacob Abel. Abel added acetic acid and pyridine to samples of amorphous (regular) insulin and, upon leaving the samples overnight, crystals of insulin formed. These crystals could be used in an X-ray diffractometer! Only one problem, diffractometers had not yet advanced enough for crystals of this complexity to be analysed...

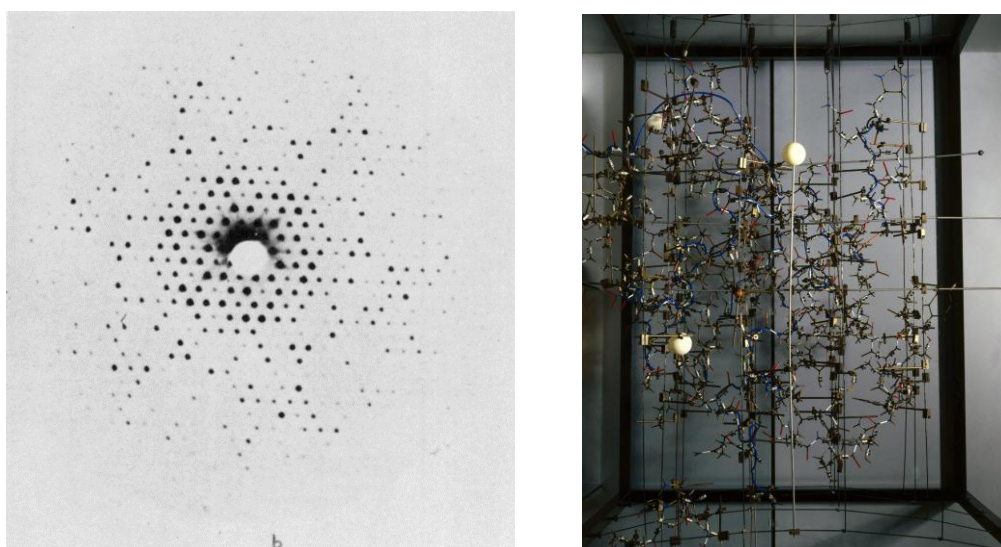


Crystals of Rhombohedral Insulin; LTR: The 1926 Observation, NASA Microgravity Crystallisations

Hodgkin took up the study of insulin next, following a gift from Robert Robinson in 1934, but was unable to make much progress until diffractometers had improved by 1969 - at which point she joined a group of young researchers to obtain both photographs and analysis.

Dorothy Hodgkin had first photographed insulin in 1935 with the lower-resolution diffractometers of the time and thus obtained a clearer photograph of insulin in record time. But, here comes problem two - analysis. It was common at the time to use models to construct possible arrangements of the structure which could then be proved or disproved using stereochemical theory, mathematical frameworks developed to analyse the stability of stereoisomers - versions of molecules with the same atoms but different spatial arrangements. However, modelling such a complex molecule had never been done before - who knew how difficult and long it may be?

Hodgkin and her team flitted between both models and diagrams over the course of three years to develop the best possible structure for insulin, tearing down model after model, until on one fateful day, the maths checked out - the final structure was found.



Analysing Insulin; LTR: (Lead) Insulin X-Ray Photograph, the Final Model

X-ray crystallography was first developed in 1912 and proves itself to chemists time and again as the go-to tool for chemical structure analysis; it is still used today on hundreds of active research projects - not bad for a 110-year-old science!

BIOLOGY

WHEN DINNER SAYS NO!

NOM, NOM, NOM! What's that sound? A ravenous monarch caterpillar, munching its way through a milkweed leaf. As you know from GCSE biology, plants are 'producers' in an ecosystem, providing energy for higher trophic levels. Therefore, animal and insect consumption of plants is an acceptable occurrence, necessary even - but only to an extent. Uncontrolled herbivory leads to a major loss of resources, limiting a plant's growth and reproduction, therefore plants fight back! These 'docile' organisms secretly have an arsenal at their command, from morphological to chemical, ecological to communicational.

Simply put, structural (or morphological) defences make a plant more difficult to eat, like an overcooked steak with no steak knife. These include waxy cuticles on stem and leaf surfaces to prevent aphid stylets from penetrating phloem sieve tube elements, spines and glochids (fine, hair-like, hooked barbs), and trichomes. Trichomes are uni- or multicellular, branched or unbranched, glandular or non-glandular (what a mouthful) appendages of plant epidermis that, among other things, physically and chemically protect against insects, secreting repellents and other toxins. These structures are pretty 'standard', but... we're not finished. Some adaptations are very interesting. For example, the apical meristem (growing point) in the Poaceae family (grasses) is found at the base of the culm or tiller (stem), below the level of grazing teeth or mowers. Thus, blades can regrow from meristematic cells undergoing mitosis, using energy (ATP and other metabolites) from stored reserves and other leaves - this is why we can have lawns as well-kept as Brad Pitt's buzz cut!



LTR: Monarch Butterfly Caterpillars Avoid Latex-Containing Vessels To Continue Eating the Leaf, Hair-Like Trichomes.

Our next step takes us into the library of chemicals at plants' proverbial fingertips to fend off would-be Sunday lunchers. Have you ever considered the lovely aromas of mint or the piquant heat of your chilli con carne (or chilli sin carne if you prefer) and thought how kind plants are to provide us with these flavours, they must be so generous - **WRONG!** Plants are selfish little things: many of the chemicals that spice up our lives are synthesised to defend themselves against being eaten - didn't work for us then did it? On a serious note, these are usually phytochemicals or 'secondary metabolites', produced by but not directly involved in photosynthesis. There are two general ways these are activated - the same chemical may be activated differently in different plants or even organs of the same organism. Phytoanticipins are stored as inactive forms, in tissues or the vacuole for example, and are activated by hydrolytic enzymes whereas phytoalexins are synthesised de novo after pathogen or insect attack; it's the difference between stockpiling long life or buying fresh milk.

Plants often combine approaches to achieve a synergistic effect. Nitrogen-containing alkaloids, such as caffeine or the opium synthesised by the *Papaver* (poppy) genus, disrupt and overload the nervous system of insects and other organisms, obviously including us. Another set of compounds are terpenoids, derived from a five-carbon isoprene unit including digitalin from foxgloves (*Digitalis*). Can you guess what this does? Bingo! Digitalin wreaks havoc on the cardiac system. Other examples include latex, this toxin also enacts a mechanical barrier in *Asclepias*, milkweed [see above]; or the volatile oils of the *Lamiaceae*, mint, family namely menthol, menthone and others.

Finally, tannins and flavoursome flavonoids, like the strong antioxidant catechins found in teas, are part of the large family of phenolics so-called for their 6-carbon benzene ring and hydroxyl group. Tannins are bitter and astringent, having deleterious effects on phytophagous (fancy word for ‘plant-eating’) insects through the chelation of metals (forming a metal complex) and impeding protein digestion. All this reduces plants’ nutritive value, dissuading many herbivores. If only cake contained tannins, diets would be a breeze!

Okay, that's the hardcore chemistry done - well, hardcore for biologists. Now we can talk about the fun stuff! Physicists, put away the quantum computers and get a load of this: Some plants exploit mutualistic ecological defences to protect themselves - for no reason other than laziness! *Tut* Anyway, wounding a plant and chemicals in caterpillar saliva, for example, trigger signal transduction pathways that release volatile chemical ‘warnings’ for other plants to prepare, such as by synthesising tannins. Similarly, this can release attractants for parasitoid wasps which lay their eggs inside the herbivorous insect, and the larvae subsequently consume it from the inside - gross!

The most exciting defences involve the myrmecophilous bullhorn acacias. Their nectaries secrete nectar (shock!), Beltian bodies provide proteins and lipids, and their large, hollowed stipules shelter *Pseudomyrmex ferruginea* ants. Here comes some chemistry - I know I said we had finished but... Bullhorns lack the bitter alkaloids of other acacia species and rely on the aggressive nature of the ants to protect them from other ant species and herbivores, such as cattle. The ants even remove nearby seedlings to maximise the sunlight that reaches their host. Ant acacias do not develop swollen thorns etc straight away, so once the ants colonise it, the plant secretes chitinase in its nectar which inhibits invertase - the hydrolytic enzyme for sucrose found in most other nectars - rendering the ants completely reliant on the acacia for nutrition. There you go folks, don't eat bullhorn acacia nectar or say goodbye to your Kellogg's Frosted Flakes - you've been warned!



LTR: Spindle Spines of the Bullhorn Acacia, Parasitoid Wasp Eggs on a Sphingid Caterpillar.

RECOMMENDATIONS

A-LEVEL WIDER INDEPENDENT STUDY

- 📖 **Mathematics:** The Sieve of Eratosthenes.
- 📖 **Physics:** What Are Lagrange Points?
- 📖 **Chemistry:** Zero-Calorie Glucose.
- 📖 **Biology:** What Is the Lymphatic System?

DISCRETE FURTHER READING

- 📖 **Mathematics:** *Quanta Magazine June 2024*, How the Square Root of 2 Became a Number ∞
- 📖 **Physics:** *JHU Engineering Spring 2024*, Reimagining a Quieter Drone ∞
- 📖 **Chemistry:** *Periodic Issue 11*, Poisonous Tales ∞
- 📖 **Biology:** *Nature Podcast February 2024*, Audio long read: Chimpanzees are dying from our colds — these scientists are trying to save them ∞

CONTINUOUS FURTHER STUDY TOPICS

- 📖 **Mathematics:** The Möbius Strip
- 📖 **Physics:** Superconductors
- 📖 **Chemistry:** Alpha Helices
- 📖 **Biology:** Structure and Function of Trichomes

BOOK APPRAISAL

Watson, J.D. (1968). *The Double Helix: A Personal Account of the Discovery of the Structure of DNA*. [online] U.S.: Atheneum Press. Available at: archive.org/details/tails/james-watson-the-double-helix ∞.

MONTHLY PROBLEM

Solve for x :

$$\frac{9^{-1}}{81^x} = 9^{x^2}$$

Solution next month.

Last month's solution:

$$x = 2$$

CREDITS

SOURCES & PROVENANCE

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EDITORS & WRITERS

✉ **NSR:** Mr. N. Skinner (Editor & Lead Writer)

✉ **OCN:** Mr. O. Chapman (Biology Writer)

✉ **AHT:** Mx. A. Herbert (Content Advisor)

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CORRECTIONS

No new errors have been identified within previous issues of this newsletter. The editor can be contacted regarding errors and suggestions via the email skinner-science-spectrum@magna-aspirations.org .